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Criminalistics

The use of error rates to predict and characterize the efficacy of presumptive field tests

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Abstract

From a forensic perspective, a presumptive test, one which indicates the presence or absence of a certain target material such as blood, is an invaluable tool. Among these tests, there are different specificities, sensitivities, and shelf lives. The accuracy of a test is an algebraic combination of the specificity and sensitivity of the test. Each test has limitations as given by its false positive and false negative rates. The aim of this study was to illustrate how the false positive and false negative rates are to be properly determined using a simulation study for the phenolphthalein test. New presumptive tests must be properly evaluated/validated through testing of commonly encountered household items and other potentially probative items usually found at crime scenes, however, the makeup of test sets must appropriately capture *all* error rates. In order to correctly use these results when the test is applied to an unknown sample recovered at a crime scene, the error rates cannot be applied directly to estimate whether or not the sample is actually the analyte of interest. In a validation study, the forensic scientist calculates the false positive rate as the p(Positive Reaction|Blood), whereas at the scene, the crime scene investigator wishes to determine the p(Blood|Positive Reaction). All crime scene investigators need to ensure that the conditional is not transposed when interpreting such results. Furthermore, this work provides a model for the assessment of a multiple test diagnostic system intended for investigators.

KEYWORDS
Bayesian interpretation, error rates, presumptive test, probability, sensitivity, specificity

Highlights

- A change in presumptive test validation is needed to further understand test performance and error.
- Equations for the assessment of a multiple test diagnostic system are given for investigators.
- A guide to calculate error rates for presumptive tests is outlined to define the p(analyte|+).

1 | INTRODUCTION

Presumptive tests are generally conducted at crime scenes to provide the investigator with preliminary information regarding the object or material being tested. According to Bell [1], a presumptive test "…does not provide definitive identification; rather, it provides information useful for directing further analysis." The purpose of a presumptive test is to provide an inference about the nature of the substance tested (e.g., presence of blood) for sample collection and whether further testing should be undertaken (a confirmatory test) in the lab. Ideally, these tests should be sensitive, specific, and robust enough to be used at crime scenes. Robustness, also known as ruggedness, is the degree to which results are unaffected by changes in the experimental conditions (e.g., small changes in temperature) which characterizes reliability during normal usage [2]. An optimal presumptive test should also be easily prepared and used, have an extended shelf-life, and be nontoxic to the user [3–5].

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Presumptive tests are qualitative in nature as the tests are generally based on chemical or physical properties, such as reactivity, to achieve a binary identification result. For qualitative methods, precision cannot be expressed as a standard deviation or relative standard deviation but may be expressed through true and false positive rates, and true and false negative rates [2,6]. During a validation study, an analyst generally collects samples, which reflect the materials commonly encountered at a crime scene to evaluate the specificity of the test. These materials should include both bodily fluids and nonbodily fluids. Additionally, some studies include samples which are known to provide false positive results [4,7,8]. The validation study will collect data on the performance of a test and calculate the various error rates. As an example, the true positive probability, often referred to as the true positive rate (tpr) is calculated as the p(+ |B). However, at the scene, the crime scene investigator (CSI) wishes to determine the p(B| +), where +represents a positive reaction and B represents the presence of blood. Given this, the generally published error rates for a specific presumptive test cannot be applied directly to estimate whether or not the sample is actually blood.

1.1 | Sensitivity

Diagnostic sensitivity is the proportion of samples with a condition (presence of analyte), that is, concentration above cutoff, which have positive qualitative test results [2]. Sensitivity refers to the ability of a test to detect the presence of an analyte given the true presence of the analyte. Differences in sensitivities have been reported for presumptive blood tests. These differences may be attributed to differences in reagent concentrations, methods of sample or reagent preparation, and in the type of material containing the blood [3,9]. Discrepancies may also be due to the presumptive test reagents being added directly to a dilute blood solution, thereby diluting the reagent. In other cases, the dilute blood solutions are dried first and then tested with full-strength reagents [9]. Grodsky et al. [9] also report that dried bloodstains are not comparable with the same amount of blood dissolved in a solution.

An important factor in determining the sensitivity of a test is the dilution factor or limit of detection (LOD) [3]. The LOD is the lowest detectable level of analyte distinguishable from the background by the test, generally given as three standard deviations from the blank value (or blank signal) [2]. However, several studies [4,5,8–10] appear to use the terms sensitivity and LOD interchangeably, when in fact the terms are merely related [2,11]. LOD is the lowest level at which an analyte is detectable, whereas sensitivity relates to the slope of the calibration curve [11]. As an example, the interrelationship for a linear calibration curve ($y = a + xb$) where y_0 is the average signal of the zero calibrator (background subtracted zero), and σ is the standard deviation of the zero calibrator [11]:

$$\text{LOD} = \frac{(y - a)}{b} = \frac{(y_0 + 2\sigma - a)}{b} \quad (1)$$

Furthermore, sensitivity or the true positive rate is calculated by the number of true positives (tp) divided by the total positives (AP) using the following equation:

$$\text{tpr} = \frac{\text{tp}}{\text{AP}} = \frac{\text{tp}}{\text{tp} + \text{fn}} = 1 - \text{fnr} \quad (2)$$

An illustration of the difference is also given in Figure 1. The frequency of positive test results for positive samples is given for a range of sample concentrations. Here, the LOD is shown where the first positive result is generated. The sensitivity cut-off for the test is shown where all positive samples are correctly classified in the qualitative system; the sample concentration at which the method can be applied reliably. The space between the vertical lines is where a proportion of the tests give a positive result, however, there is not agreement. As traditionally done, assessing the LOD as the sensitivity of the test introduces error to the test in the form of false positives.

1.2 | Specificity

Diagnostic specificity is the proportion of samples without a condition (analyte not present), that is, concentration below cutoff, which have negative qualitative test results [2]. The specificity of a test describes the ability of the test to detect the absence of an analyte in instances where the analyte is truly absent. Tests for specificity of presumptive tests have included changing substrates, adding material and chemicals to the bloodstains, and testing to see whether the reagents will react with substances other than blood [3,4,12–16]. Grodsky et al. [9] state that presumptive blood tests with a degree of interference effectively prevent their use as a test for the presumptive identification of blood and therefore further examination is required. A comparison between sensitivity and specificity based on the results of the test is given in Figure 2.

The specificity or the proportion of samples without the analyte/target is calculated by the number of true negatives (tn) divided by the total negatives (AN) using the following equation:

$$\text{Specificity} = \frac{\text{tn}}{\text{AN}} = \frac{\text{tn}}{\text{fp} + \text{tn}} \quad (3)$$

1.3 | Selectivity

Selectivity is a concept highly related to specificity. The selectivity of a qualitative method is the extent to which other analytes than the target interferes with the analysis [2]. This feature is fundamental to the error rate of the method; the ability of the test to produce results not influenced by matrix effects. A test that is selective in nature provides a more accurate sample classification as only the analyte of interest will give a positive indication.

1.4 | Validation of presumptive tests

Method validation is the process used to demonstrate that an analytical procedure for a specific test is suitable and fit for its intended purpose. The results from a method validation can be used to assess the reliability

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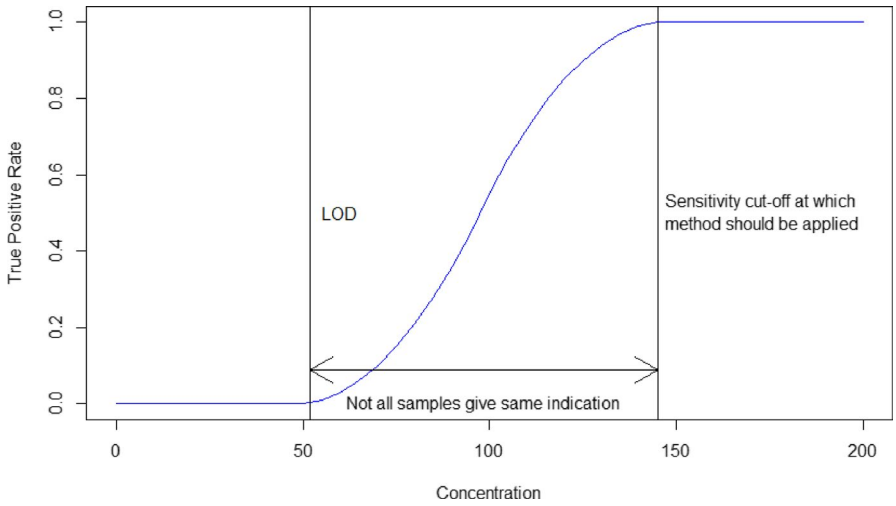


FIGURE 1 Frequency of true positive samples which yield a positive test reaction as concentration increases. Note that this example uses arbitrary concentrations to illustrate the difference between LOD and sensitivity [Color figure can be viewed at wileyonlinelibrary.com]

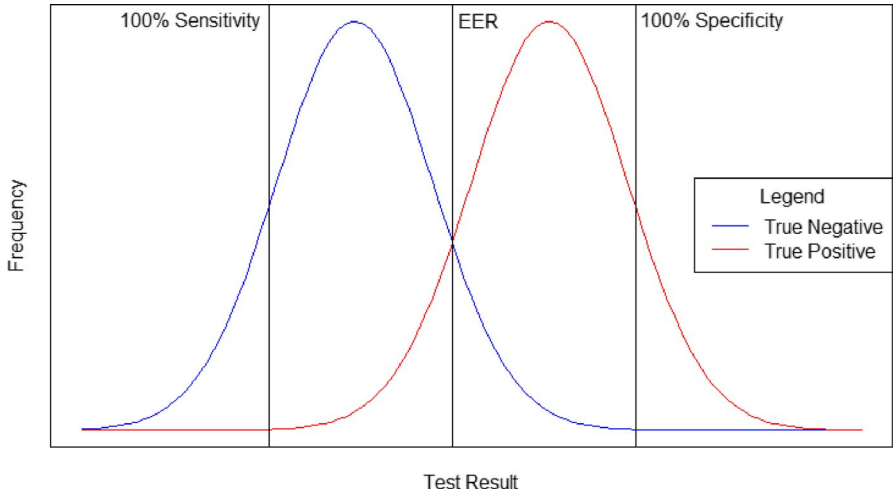


FIGURE 2 Comparison between the frequency of true positive and negative results. Here, 100% sensitivity is given where there are no missed true positive samples, and 100% specificity is given where there are true negative samples. The equal error rate (EER) is also shown [Color figure can be viewed at wileyonlinelibrary.com]

and consistency of the test and determine whether the method fulfills requirements for the intended use prior to implementation [17]. An often-omitted concept in the validation of a new presumptive test is the sample makeup of the validation set. Generally, such testing is performed by an analyst in a blind fashion. The preparer creates the samples and withholds the ground truth, while the validator conducts the analysis and provides the results, usually in the form of a positive or negative reaction for the samples. Additionally, an assessment can be performed to establish the LOD for the presumptive test [2].

Questionably, should blood samples below the LOD be included in the test and, if so, how many? The exclusion of such samples removes the ability of the validator to determine the fnr for the test. This is especially of concern because the investigator does not know the concentration of a target sample at a crime scene, nor the concentration of any other trace elements beneath/mixed with the sample. There are also combinations of blood and other reagents which may react with one another and result in a false negative reaction (e.g., blood and bleach). Similarly, how many false positive samples should be included in the validation set? If examiners wish to assess the performance of one reagent versus another using the fnr and fpr, then the reagents should have been tested on the same validation sample set for consistency and fair comparison.

1.5 | Aims of the study

The aim of this study is to illustrate how the false positive (fpr) and false negative rates (fnr) should be determined and used with regard to presumptive tests. A presumptive blood test should be utilized when the context and physical properties do not clearly indicate blood (blood will be used as the analyte throughout this paper, but the concept is the same for all analytes). Interpretation of presumptive tests is significant to the investigator because in many cases involving suspected trace samples of blood, the investigator would not collect the stain given a negative indication by the test. Such situations place an increased importance on the sensitivity and specificity of the presumptive test employed.

2 | MATERIALS AND METHODS

2.1 | Test sets

A simulation study was performed to assess the Kastle–Meyer or phenolphthalein (one step) test for blood. The false positive sample list by Sutton [7] was used. The fpr is given as:

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$$\text{fpr} = \frac{\text{fp}}{\text{fp} + \text{tn}} = \frac{\text{fp}}{\text{AN}} \quad (4)$$

In the true negative data set, there were 193 samples, of which 13 gave a false positive reaction (of varying degrees) for blood. Using these data, a simulation of 10,000 tests was conducted. For each iteration of the simulation, a random number was generated (1–1000) which represented the number of true blood samples. The "actual" blood sample set was generated by randomly selecting a sample set size (*S*) and then selecting a random proportion (p_LOD) of samples that were below the LOD. Thus, for that set the fnr is given by:

$$\text{fnr} = \frac{\text{p_LOD} \times S}{S} \quad (5)$$

The implementation of LOD enabled the generation of both true positive and false negative samples within the simulation. Next, a random number was generated (1–193) which represented the number of true negative or other samples randomly chosen from the Sutton dataset. In each random sample, the test results were counted, allowing for the calculation of thesample tpr, tnr, fpr, and fnr for each iteration. An example of one iteration of the method is illustrated in Figure 3. This process was repeated for all 10,000 tests. The simulation source code is available in Supporting Information.

3 | RESULTS AND DISCUSSION

The potential variability in error rates was evaluated by sample size of the test/validation set using the rates from the 10,000 simulations. These fpr data were plotted as a function of the sample size in Figure 4. In Figure 4, the dots represent the simulated fpr. The fitted curves model the rates for all possible sample set sizes. The dashed line indicates the actual *fpr* of the set $\left(\frac{13}{193} \cong 0.067\right)$. Statistically, a fpr of 1.0 is possible in sample sets of size 1–13, if all of the false positive samples are chosen. In smaller sets, an fpr of zero is also encountered often. In the simulation, a fpr of zero was not observed at a set size of approximately 75. After the set of 75, the simulated fpr tended toward the overall fpr of the false positive set. Overall, the curves in Figure 4 provide a range of fpr that can be encountered in a typical validation study. The results also support the assertion that a standard sample test set should be developed to provide a fair assessment for validation studies of presumptive tests. If any generalized validation set was chosen, then a similar tendency would be seen. During all of these simulations, the test was assumed to be 100% effective.

In Figure 5, the dots represent the simulated tpr. The fitted curves represent the rates for all possible sample set sizes (*n*), as given by:

$$\text{tpr}(x) = \frac{i}{S}, \text{where } i = 1, \ldots, n, \text{and } S = \text{sample set size} \quad (6)$$

Since no minimum fnr was specified, the tpr in this simulation can assume any value between zero and one. If an fnr was chosen, for example, 0.2, then tpr = 0.8.

3.1 | Evaluating a new presumptive test

Forensic scientists generally accept that some form of validation is required for a test before the test is applied. The main purpose of validation is to estimate the reliability of the test under consideration. Quality systems within labs and accrediting bodies (e.g., ANAB ISO/IEC 17025) require the determination of measurement uncertainty. Error in this context is defined as the error within a method. This error is the inability of a test to correctly classify the outcome of the test and may be thought of as the deviation of the sensitivity and specificity from unity.

To illustrate how this can be done, the process of evaluating a new "presumptive" test kit for blood is presented. After selection of a realistic set of samples encountered by a CSI, suppose that the test was determined to have an fnr of 0.10 and an fpr of 0.05. After the completion of the validation phase, the test was deployed for use on crime scenes. The CSI identifies a stain of interest and performs the test on the sample. The test indicates a positive result. Given this test, what is the probability that the sample is actually blood? CSIs are interested in determining p(B| +), which requires the use of Bayes theorem. It is already known that:

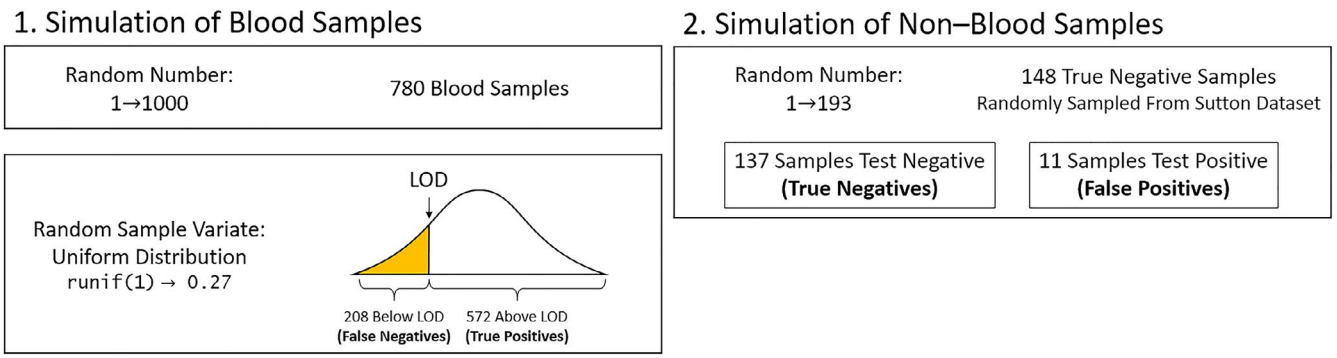


FIGURE 3 An example of one simulation of a presumptive test and how samples were randomly generated [Color figure can be viewed at wileyonlinelibrary.com]

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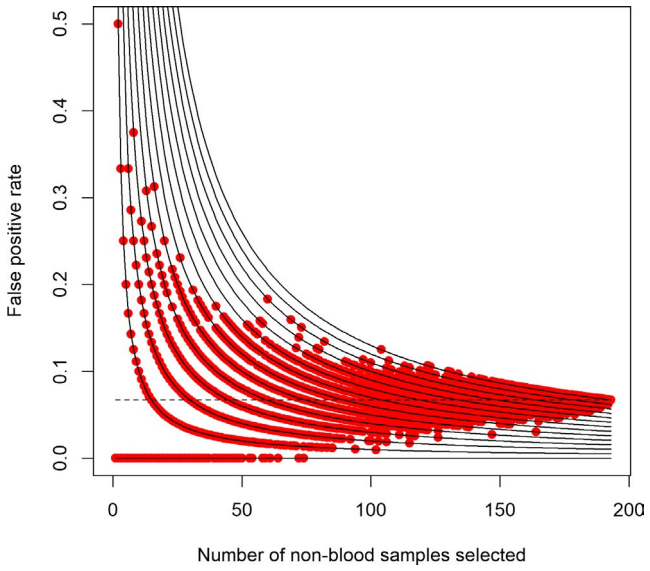


FIGURE 4 The fpr of randomly selected samples from the actual negative set as a function of the sample set size [Color figure can be viewed at wileyonlinelibrary.com]

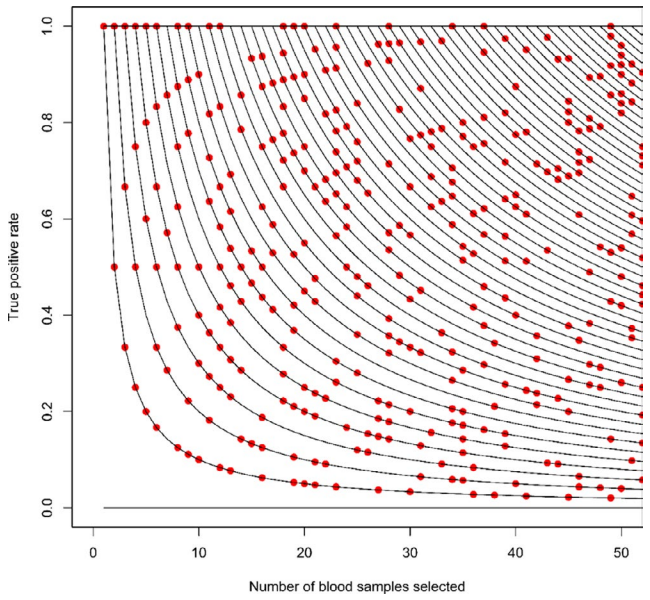


FIGURE 5 The tpr of randomly generated false negative samples in a randomly sized truly positive set [Color figure can be viewed at wileyonlinelibrary.com]

$$\begin{aligned} \text{fpr} &= p\left(+|\overline{B}\right) = 0.05, \text{ and thus } \text{tnr} = 1 - \text{fpr} = 0.95 \\ \text{fnr} &= p\left(-|B\right) = 0.10, \text{ and thus } \text{tpr} = 1 - \text{fnr} = 0.90 \end{aligned} \tag{7}$$

These rates for the test are also shown in Table 1 in relation to one another.

When this information is entered into Bayes theorem, the following is true:

$$p(B|+) = \frac{p(+|B) \times p(B)}{p(+)} = \frac{p(+|B) \times p(B)}{p(+|B) \times p(B) + p\left(+|\overline{B}\right) \times p\left(\overline{B}\right)} \tag{8}$$

TABLE 1 Rates for the sample presumptive test being evaluated

	Test result	
	+	–
Ground truth		
+	0.95	0.05
–	0.10	0.90

This calculation is also known as the positive predictive value for the test; the probability that the analyte is present when the test is positive [2,18]. From (8), it is evident that p(+ |B) or the tpr is not equal to p(B| +). Interchange of these conditional probabilities is known as transposition of the conditional—a significant problem seen in the interpretation of statistical evidence for criminal trials [19,20]. For this example, a value of 0.01 is used for p(B), thus (8) yields:

$$p(B|+) = \frac{0.90 \times 0.01}{0.90 \times 0.01 + 0.05 \times 0.99} = 0.15 \tag{9}$$

In Figure 6, Bayes theorem has been used to calculate the p(B| +) for four sets of fnr and fpr as a function of p(B). It is clear that p(B) has a marked effect on p(B| +) and that the utility of any presumptive test cannot be separated from p(B). Therefore, it is incumbent upon every CSI, and every DNA analyst, to ensure that they understand the p(B) for the case being considered.

3.2 | Multiple test system

In a two-test system, two tests are applied sequentially to a specific sample. In a simple system, these tests are independent in terms of reaction chemistries. As an example, amido black reacts with the

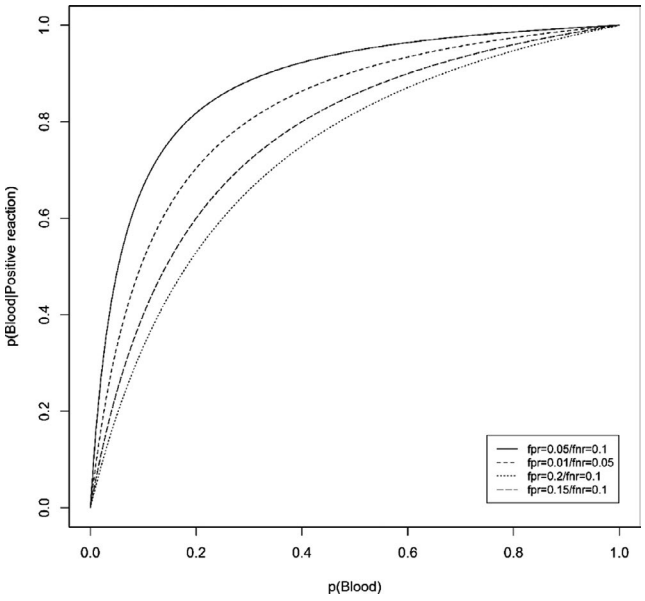


FIGURE 6 Representation of the p(B| +) with given fpr and fnr as a function of the p(B).

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proteins in blood [21] and phenolphthalein is dependent on a redox reaction to convert the iron within hemoglobin [7].

In the two-test system, $Test_1$ and $Test_2$ are applied sequentially. Given the various error rates, the number of tp, tn, fp, and fn can be determined using the number of actual positives and negatives (see Equation 10). For the two-test system where AP = the number of actual positive sample in the test set and AN = the number of actual negative samples in the test set:

$$\begin{aligned} tp_1 &= tpr_1 \times AP \\ tn_1 &= tnr_1 \times AN \\ fp_1 &= AN - tn_1 \\ fn_1 &= AP - tp_1 \end{aligned} \tag{10}$$

Figure 7 demonstrates the flow of a sample through two independent tests. After $Test_1$ is completed, there are two groups which contain the positive and negative samples. The negative group will contain both tn_1 and fn_1 samples. The true negative samples are not of concern since those do not contain the analyte of interest, whereas the fn_1 samples do contain the analyte of interest. This is of great significance because the CSI does not collect negative samples; these fn_1 samples are then lost for further analysis. The positive samples, both the tp_1 and fp_1, are then reanalyzed using $Test_2$ because the analyst has no information other than that the presumptive test yielded a positive result. $Test_2$ yields the same type of result. These are given by Equation (11).

Overall, for the two tests, the following is true:

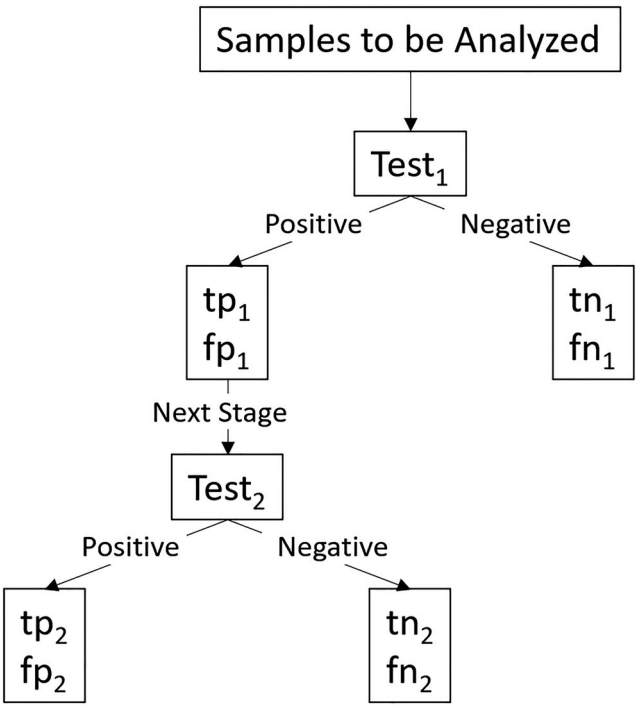


FIGURE 7 Flow chart demonstrating the use of two independent tests in sequence

$$\begin{aligned} tp_0 &= tp_2 \\ fp_0 &= fp_2 \\ fn_0 &= fn_1 + fn_2 \\ tn_0 &= tn_1 + tn_2 \end{aligned} \tag{11}$$

The overall rates can be further described as:

$$\begin{aligned} tpr_0 &= \frac{tp_2}{tp_2 + fn_1 + fn_2} \\ fpr_0 &= \frac{fp_2}{fp_2 + tn_1 + tn_2} \\ fnr_0 &= \frac{fn_1 + fn_2}{fn_1 + fn_2 + tp_2} \\ tnr_0 &= \frac{tn_1 + tn_2}{tn_1 + tn_2 + fp_2} \end{aligned} \tag{12}$$

It is necessary to define the number of positives and negatives for each of the two test steps:

$$\begin{aligned} tp_2 &= tpr_2 \times tpr_1 \times AP \\ fp_2 &= fpr_2 \times fpr_1 \times AN \\ fn_2 &= fnr_2 \times tpr_1 \times AP \\ tn_2 &= tnr_2 \times fpr_1 \times AN \end{aligned} \tag{13}$$

Subsequently, Equation (13) can be substituted into Equation (12) to yield:

$$\begin{aligned} tpr_0 &= \frac{tpr_1 \times tpr_2}{tpr_1 \times tpr_2 + fnr_2 \times tpr_1 + fnr_1} \\ fpr_0 &= \frac{fpr_1 \times fpr_2}{fpr_1 \times fpr_2 + tnr_2 \times fpr_1 + tnr_1} \\ fnr_0 &= \frac{fnr_1 + tpr_1 \times fnr_2}{fnr_1 + tpr_1 \times fnr_2 + fpr_1 \times tpr_2} \\ tnr_0 &= \frac{tnr_1 + tnr_1 \times fpr_1}{tnr_1 + tnr_1 \times fpr_1 + fpr_1 \times fpr_2} \end{aligned} \tag{14}$$

These principles can be further extended to encapsulate an *n*-test system. The expansion of (13) allows for the generalization of the overall rates for *n* consecutive tests to be:

$$\begin{aligned} tpr_{overall} &= \prod_{i=1}^{n} tpr_i \\ fpr_{overall} &= \prod_{i=1}^{n} fpr_i \\ fnr_{overall} &= \sum_{i=1}^{n} fnr_i \prod_{j=1}^{n-1} tpr_j \\ tnr_{overall} &= \sum_{i=1}^{n} tnr_i \prod_{j=1}^{n-1} fpr_j \end{aligned} \tag{15}$$

3.2.1 | Example calculations for a three-test system

Consider a scenario where $S = 200$ samples are being tested. Of the samples, 100 are truly positive and 100 are truly negative. Three independent tests are performed (all with different reaction chemistries) in sequence. Suppose $Test_1$ has the following rates: $tpr_1 = 0.85$; $fpr_1 = 0.10$; $fnr_1 = 0.15$; and $tnr_1 = 0.90$. Given this information the test would yield the following results:

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$N_0 = 200$
$+ = 100$
$- = 100$
Let $+ = tp_0$ and $- = tn_0$

Ground truth	Test 1 +	Test 1 −
+	85	15
−	10	90

$$tp_1 = tpr_1 \times tp_0 = 0.85 \times 100 = 85$$
$$fp_1 = fpr_1 \times tn_0 = 0.10 \times 100 = 10$$
$$fn_1 = fnr_1 \times tp_0 = 0.15 \times 100 = 15$$
$$tn_1 = tnr_1 \times tn_0 = 0.10 \times 100 = 10$$

Of the original 200 samples, only 95 remain as the fn and tn samples would be discarded after $Test_1$. The samples which tested positive would then be re-evaluated using $Test_2$. Suppose $Test_2$ has the following rates: $tpr_2 = 0.95$; $fpr_2 = 0.25$; $fnr_2 = 0.05$; and $tnr_2 = 0.75$. Given this information the test would yield the following results:

$N_1 = 95$
$+ = 85$
$- = 10$

Ground truth	Test 2 +	Test 2 −
+	80.75	4.25
−	2.50	7.50

$$tp_2 = tpr_2 \times tp_1 = tpr_2 (tpr_1 \times tp_0) = 0.95 \times 0.85 \times 100 = 80.75$$
$$fp_2 = fpr_2 \times fp_1 = fpr_2 \times (fpr_1 \times tn_0) = 0.25 \times 0.10 \times 100 = 2.50$$
$$fn_2 = fnr_2 \times tp_1 = fnr_2 \times (tpr_1 \times tp_0) = 0.05 \times 0.55 \times 100 = 4.25$$
$$tn_2 = tnr_2 \times fp_1 = tnr_2 (fpr_1 \times tn_0) = 0.75 \times 0.10 \times 100 = 7.50$$

The positive samples would then proceed to $Test_3$. The rates for $Test_3$ are: $tpr_3 = 0.90$; $fpr_3 = 0.20$; $fnr_3 = 0.10$; and $tnr_3 = 0.80$. Another iteration of testing on the remaining 83 samples would yield:

$N_1 = 83$
$+ = 81$
$- = 2$

Ground truth	Test 3 +	Test 3 −
+	72.675	8.075
−	0.500	2.000

$$tp_3 = tpr_3 \times tp_2 = tpr_3 (tpr_2 \times tp_1) = tpr_3 (tpr_2 (tpr_1 \times tp_0)) = 0.90 \times 0.95 \times 0.85 \times 100 = 72.675$$
$$fp_3 = fpr_3 \times fp_2 = fpr_3 \times (fpr_2 \times fp_1) = fpr_3 (fpr_2 \times (fpr_1 \times tn_0)) = 0.20 \times 0.25 \times 0.10 \times 100 = 0.500$$
$$fn_3 = fnr_3 \times tp_2 = fnr_3 \times (tpr_2 \times tp_1) = fnr_3 (tpr_2 (tpr_1 \times tp_0)) = 0.10 \times 0.95 \times 0.85 \times 100 = 8.075$$
$$tn_3 = tnr_3 \times fp_2 = tnr_3 (fpr_2 \times fp_1) = tnr_3 (fpr_2 (fpr_1 \times tn_0)) = 0.80 \times 0.25 \times 0.10 \times 100 = 2.000$$

Calculation of the overall error rate of performing the three tests outlined can be achieved using (15), which correspond to the values output by the above tests:

$$tpr_{overall} = \prod_{i=1}^{n} tpr_i = 0.85 \times 0.95 \times 0.90 = 0.72675$$
$$fpr_{overall} = \prod_{i=1}^{n} fpr_i = 0.10 \times 0.25 \times 0.20 = 0.00500$$
$$fnr_{overall} = \sum_{i=1}^{n} fnr_i \prod_{j=1}^{n-1} tpr_j = 0.15 + (0.05 \times 0.85) + (0.10 \times 0.85 \times 0.95) = 0.27325$$
$$tnr_{overall} = \sum_{i=1}^{n} tnr_i \prod_{j=1}^{n-1} fpr_j = 0.90 + (0.75 \times 0.10) + (0.80 \times 0.10 \times 0.25) = 0.99500$$

The implications of these equations are that the investigator or examiner can apply tests in sequence and be able to determine the overall efficacy of the sequence. Admissibility requirements under the Daubert standard require the "known or potential error rate" of the methodology to be presented [22]. When using multiple tests, the above can be used to calculate the error rates for the entire methodology performed.

3.3 | A Bayesian assessment of test performance

The true positive rate of a test system consisting of two independent tests is, at best, the true positive rate of the first test. In this instance, the false negative rate is important since all samples which test negative will not be collected; this rate has a direct impact on both tests. The true negative rate of the second test will reduce the number of false positives since all true negatives have been correctly rejected in the first test. Improvement of tnr_1 will reduce the number of unnecessary future tests. If $tpr_2 \to 1$, then $tpr_0 \to tpr_1$ and $fnr_0 \to 0$. If $tnr_2 \to 1$, then $tnr_0 \to (1 - tnr_1)$ and $fpr_0 \to 0$. These rates within a two-test system can be illustrated using a simple scenario. The performance of the two presumptive tests, viz., $Test_1$ and $Test_2$ is given in Table 2. Since the upper limit for the tpr_0 is tpr of $Test_1$, then $tpr_0(0.81) \leq tpr_1(0.85)$.

Additionally, a Bayesian network was constructed using the Netica™ application [23] by Norsys™ to illustrate the application of $Test_1$ and $Test_2$ as outlined in Table 2. Figure 8 shows the Bayesian network and the conditional probability tables for each test node. The network is a graphical structure which represents the joint

TABLE 2 Two test performance

	$Test_1$	$Test_1$	$Test_1$ and $Test_2$
tpr	0.85	0.95	0.81
tnr	0.90	0.75	0.98

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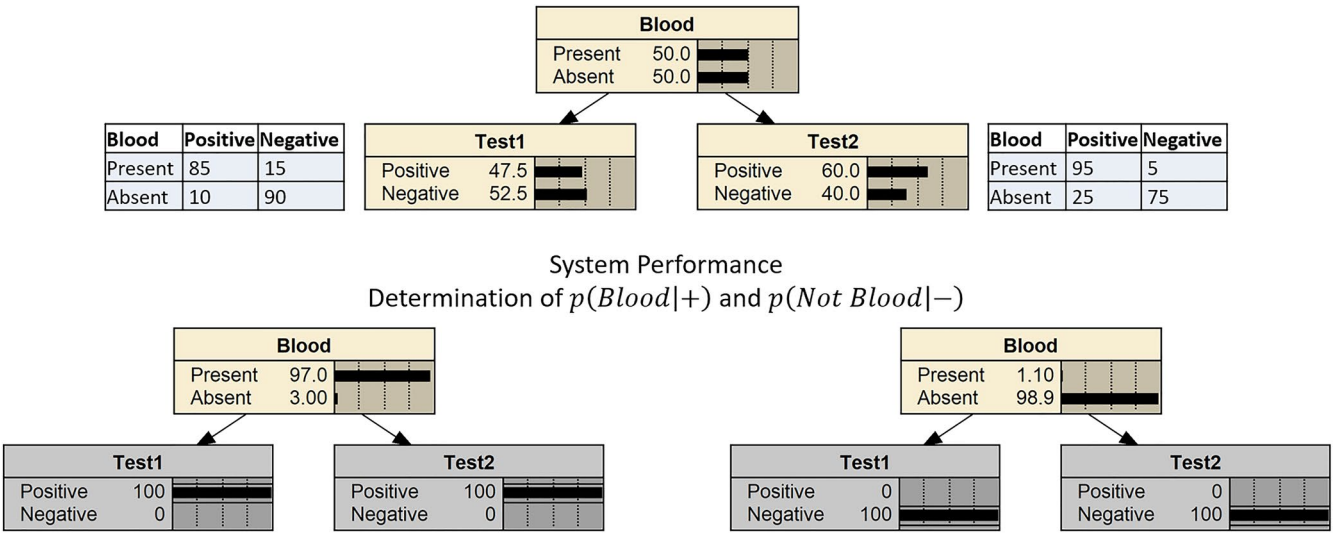


FIGURE 8 Bayesian network which depicts the two-test system of $Test_1$ and $Test_2$. The network is shown on top with the conditional probability table for each test node. Shown bottom is the performance of the system (two consecutive tests) in determining the p(B| +) and the $p(\overline{B}|-)$ [Color figure can be viewed at wileyonlinelibrary.com]

relationships between the tests and the resultant influence of the test result on the probability of blood. Also shown are the calculations of p(B|Result) through the selection or instantiation of the node condition (positive or negative result) for both nodes (tests). The resulting conditional probability distribution is then given for the "Blood" node in the network. Note that the result of both tests being either positive or negative does not result in certainty that the analyte being tested is blood. This information is actually what is sought by the investigator, the p(B|Result).

This network can also be used to illustrate the importance of the application sequence in which the presumptive tests are applied. While Bayes' theorem is invariant to the order in which multiple items of evidence are processed since the prior odds would update given any new information, the significance of the test outcome can be illustrated with this network. Figure 9 shows the application of the two tests which result in nonagreement. In these cases, the first test would have to result in a positive result for blood, therefore the testing sequence would continue. The subsequently applied test provided conflicting information to the examiner, returning a negative result. In these cases, the "Blood" node returns different probabilities based upon which test was performed first. The importance of test sequence for crime scene investigators is illustrated by the increased probability of blood being present given that $Test_2$ was performed first.

Another measure of the effectiveness for a diagnostic test is the diagnostic odds ratio (DOR). The DOR is the ratio of the odds of the test being positive given the presence of the analyte relative to the odds of the test being positive in instances where the analyte is not present [2,18]. This metric combines the concepts of sensitivity, specificity, and likelihood ratios into a single value. The positive and negative likelihood ratios are based on the sensitivity (Equation 2) and specificity (Equation 3) of the test. The likelihood ratios are then used to calculate the DOR as follows [19]:

$$\begin{aligned} LR+ &= \frac{\text{sensitivity}}{(1-\text{specificity})} = \frac{p(+|B)}{p(+|\overline{B})} \\ LR- &= \frac{(1-\text{sensitivity})}{\text{specificity}} = \frac{p(-|B)}{p(-|\overline{B})} \\ DOR &= \frac{LR+}{LR-} \end{aligned} \tag{16}$$

The standard error for range estimates and significance testing can be calculated by the following [19]:

$$SE(\ln(DOR)) = \sqrt{\frac{1}{tp} + \frac{1}{fp} + \frac{1}{fn} + \frac{1}{tn}} \tag{17}$$

The DOR was calculated for each of the 10,000 simulated tests to determine the DOR as a function of *n* samples. Since a range of sample counts (*n*) was used and there were zeros present for instances of fp, fn, and tn, transformation was necessary with the simulated tests. If a contingency table contained zeroes, the DOR will be undefined (∞ or $-\infty$). The addition of 0.5 was performed to all counts in the table; a commonly used method to calculate an approximation of the DOR [18,24]. The results are given in Figure 10, where the log(DOR) and the standard error for the DOR are given for the range of simulated samples. The standard error is included to illustrate the approximation of the DOR, which is conventionally included in instances where a contingency table contained zeroes.

The standard error for the log(DOR) of the simulated tests was shown to be of a large range when few samples are tested and narrows as the number of samples increase, which is expected. The DOR decreases over the plot (425–389); however, it is rather stable throughout the tests. The wider variation in the standard error is what is more important here, that the uncertainty of the DOR approximation changes so drastically. This would indicate that more caution should be taken in the validation of the tests.

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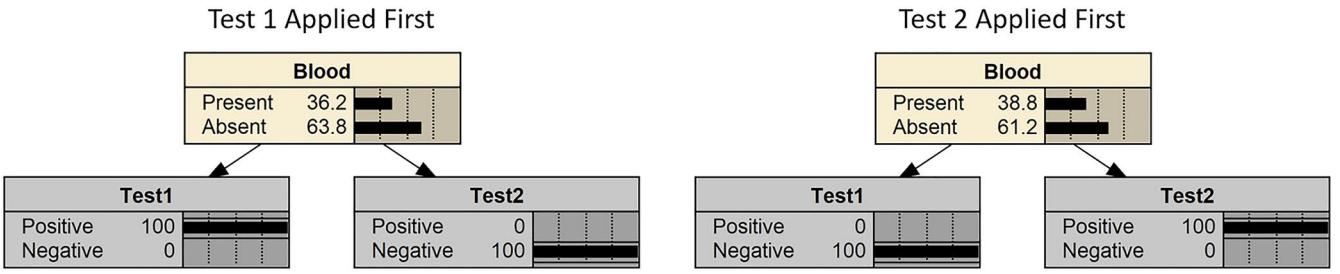


FIGURE 9 Bayesian network with a different application sequence of $Test_1$ and $Test_2$. Here, the consecutively applied tests produce results which are not in agreement, and the resultant probability for the presence of blood is given [Color figure can be viewed at wileyonlinelibrary.com]

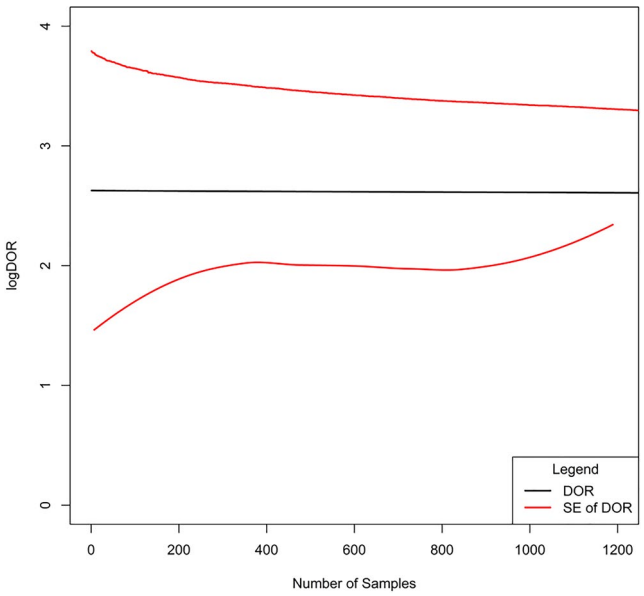


FIGURE 10 The log(DOR) and the standard error for the 10,000 simulated tests. The DOR is also given numerically for the minimum and maximum sample sizes within the test simulation [Color figure can be viewed at wileyonlinelibrary.com]

The DOR is generally used for comparing tests whenever the balance between false negative and false positive rates is not of immediate importance and is given as an odds ratio, a concept familiar to practitioners [18]. However, the balance between the true positive and false positive rates is significant within forensic science. Whenever false positives and false negatives are weighted differentially, the prevalence and conditional error rates of the test must be taken into consideration to make a balanced decision [18]. The DOR does not distinguish between the two types of diagnostic mistakes, therefore, the conditional indicators such as sensitivity and specificity still must be used. These likelihood ratios do provide utility in the comparison of the diagnostic content of multiple possible test results and for transforming those into post-test probabilities [18]. Another contribution was demonstrated in the characterization of the DOR and how wide the standard error is when few samples are tested.

3.4 | Cost associated with errors in presumptive testing

Across disciplines, various error rates are considered more important. In the case of presumptive tuberculosis testing, the risk of false positives is generally not considered for policy discussions, program targets, and indicators [25]. In these tests, a fnr is extremely concerning as the disease would go undiagnosed and attention is directed to minimize this type of error. A forensic scientist, as well as the court, preferentially attempts to reduce false positive errors because the implication is a wrongful conviction and loss of liberty [26]. In contrast, CSIs are generally more concerned about the fnr of their presumptive test because a false negative will result in the loss of that evidence (would not be collected). Biometric scientists often use the equal error rate (EER) to assess the performance of their classifiers [27]. An EER is the measure where the fpr = fnr which implies that the cost of a false positive is the same as a false negative. From the preceding information, this is not always true. The question of the relative effect, cost, or loss of one type of error versus another is now important in test selection. This cost can be expressed in monetary or relative terms.

The cost of an error is very difficult to assess when it is related to a forensic case. Simplistically, if a sample yields a false negative result, then the CSI will not collect the sample, and, if the person who deposited the sample was the actual perpetrator, then they could possibly go free; free to commit other crimes, and/or to increase the cost of the investigation. This kind of error will never get to court and is thus within the responsibility of law enforcement. A false positive may result in increased workload at the laboratory. Furthermore, a false positive during the application of a presumptive test for drug evidence could lead to the detention of an individual until confirmatory testing could be done; falsely implicating the individual.

To demonstrate the utility of an *n*-test system, an example of potential cost, or loss to an agency in the event of an error is assessed. From a cost of error or a potential loss perspective, assume that a = the cost of a false negative and b = the cost of a false positive (AP = actual positives and AN = actual negatives):

$$\begin{aligned} \text{Cost} &= a\times AP\times\left(fnr_1+fnr_2\times tpr_1\right)+b\times AN\times\left(fpr_1+tpr_2\times fpr_1\right)\\ &= a\times AP\times fnr_0+b\times AN\times fpr_0\\ &= a\times fn_0+b\times fp_0 \end{aligned} \tag{18}$$

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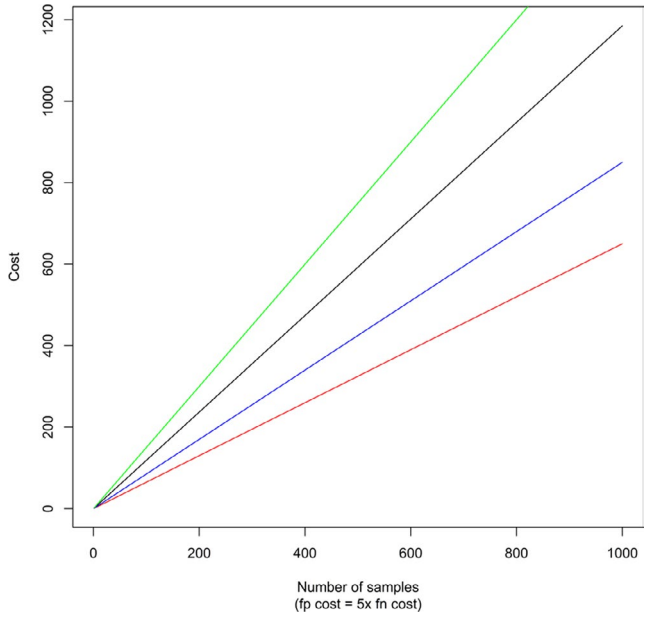


FIGURE 11 An illustration of potential cost or loss for false positives and false negatives, where the cost of a false positive is 5× that of a false negative, as a function of various sample sizes. The red curve is the cost of Test_1, the blue curve is the cost of Test_2, the green curve is the sum of the costs of Test_1 and Test_2, while the black curve is the cost of the combination of Test_1 and Test_2 [Color figure can be viewed at wileyonlinelibrary.com]

If AP = AN = n, then Equation (18) reduces to:

$$\mathrm{Cost} = n \times \left(\mathrm{a} \times \mathrm{fnr}_0 + \mathrm{b} \times \mathrm{fpr}_0\right) \tag{19}$$

An example of the cost analysis for Test_1 and Test_2 is given in Figure 11. The cost of false positives and false negatives is considered, where the cost of a false positive is five times greater than that of a false negative. This is given as a function of various sample sizes. Illustrated is the cost of each test independently, the summation of these costs, and the cost of sequential application of the tests. Evident is that the combined application cost for the tests is less than the summed cost of the tests. This is due to the change in error rates for the overall system (two tests), namely, the improvement of the tnr to 98% as shown in Table 2. In this example, the usage of an n-test system would serve to mitigate potential cost or loss to the lab given an error, either false positive or false negative.

4 | CONCLUSION

As far as the validation test sets are concerned, it is suggested that a standardized test set be adopted. Such a test set should include a dilution series of the analyte of interest, at levels below those currently assessed on existing tests. A standard number of substances yielding a false positive, covering all reaction chemistry types, must also be included. The size of both the actual negative and positive components needs to be standardized and established. At present, there is no standardized test set in the case of validating presumptive tests, rather labs are left to complete their own assessments. Ideally, a standardized protocol for validation of presumptive tests would be a contribution of the National Institute of Standards and Technology Organization of Scientific Area Committees for Forensic Science (OSAC). A uniform procedure would enable a consistent evaluation of the error rates across tests for intra-comparison, similar to the OSAC Standard for Validation of Probabilistic Genotyping Systems or the standard for Mixture Interpretation Protocols. Upon the design of such a set, a re-evaluation of all commonly used tests should be undertaken and their relative performance can be assessed. Such a standard would also allow for the assessment of new or innovative tests in both effectiveness and efficiency.

The results also indicate that background studies should be done to assess the recognition and understanding that CSIs have in regard to the p(B) for a given case or scenario due to the marked effect this has on the test. Subsequently, the implementation and use of Bayes theorem or a Bayes net allows for the determination of the p(B| +) for the CSI. This process was also demonstrated to assist in the sequence determination for multiple tests in a system. Execution of multiple tests in sequence requires careful consideration built on a proper validation of the tests to ensure that the overall error rates are fully reduced to the capabilities of the tests. Therefore, validation studies should create a dataset which ensures an evaluation of all error types, including false negatives. Additionally, the equations from this work can be utilized to ensure that an n-test system is achieving overall error rates as low as possible given the capabilities of the tests.

Different approaches for the evaluation of the qualitative tests are needed, namely, the determination of diagnostic sensitivity so that it is not mistakenly presented by the LOD. This article does not outline what the ideal composition of the test set should include but does call the need for a uniform method to attention of practitioners to ensure a quality system. More accurate evaluations of presumptive tests will further enable assessment of the costs associated with multiple tests in a system.

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